SE FINAL PROJECT SPECIFICATION

Course: Software Engineering | Semester: Sem01_2627 (Semester 1, 2026-2027) | Version 1.0

Topic: SaigonBite – Cloud-Kitchen Food Delivery Hub. **Team Size:** 2-3 students per group.

1. Background & Problem Statement

SaigonBite is a planned network of **cloud kitchens ("hubs")** in Ho Chi Minh City. Each hub cooks for several virtual brands (for example a pho brand and a banh mi brand). Customers order through one platform, kitchen staff prepare the order, and a shipper delivers it.

The platform must serve two distinct customer groups: **Local Customers** (Vietnamese phone number; MoMo, VNPay or cash) and **Foreign Tourists** (passport number and email; Apple Pay, PayPal or cash). Hubs are also unevenly loaded: crowded at lunch and dinner, idle at other times. The operator has commissioned your software engineering team to design a unified system that manages hubs and menus, calculates prices with load-aware rules, runs each order through a controlled lifecycle, and supports a multi-option payment architecture.

Educational Objectives: This project simulates a real-world software engineering contract. You will practice the complete Software Development Life Cycle (SDLC): from abstract client requirements to system modelling, UI/UX design, database structuring, coding and rigorous quality assurance testing. Like a real client, this specification **deliberately leaves some details open** (for example how a shipper is chosen, or what is refunded after a failed delivery). Spotting and resolving such gaps is part of the assessment.

2. Core Business Logic & System Features

Regardless of your technology stack, your system must implement the following actors, business rules and features. Rule IDs (BR1 to BR18) are fixed so that you can cite them in requirements, designs and test cases.

A. Actors

Actor	Description
Customer (abstract)	Generalization of the two customer types below. Places and tracks orders, pays, cancels when allowed.
Local Customer	Registers with a Vietnamese mobile number (simulated OTP). Pays with MoMo, VNPay or Cash on Delivery.
Foreign Tourist	Registers with passport number, nationality and email (simulated email code). Pays with Apple Pay, PayPal or Cash on Delivery. May enter a hotel name and room as delivery address.
Kitchen Staff	Belongs to one hub. Confirms, prepares and marks orders ready; rejects orders with a reason.
Shipper	Picks up ready orders, delivers them, confirms cash collected, or reports a failed delivery.

Actor	Description
System Admin	Manages hubs, menus, vouchers and users; sets the heavy-rain flag; views reports.

Your database must store Local Customers and Foreign Tourists in a **single schema** (for example one Users table with a role or discriminator plus a profile table per customer type) and your design documents must show the generalization.

B. Business Rules

All amounts are in Vietnamese dong (VND). Time is Vietnam local time (UTC+7). Distances are measured in km from hub to delivery address (you may store or simulate the distance).

ID	Rule	Notes
BR1	Minimum order. The item subtotal S (before any discount) must be at least 50,000. Otherwise the order is rejected.	Boundary: 49,999 / 50,000
BR2	Coverage. The delivery distance must be 8.0 km or less. Otherwise the order is rejected.	Boundary: 8.0 / 8.01
BR3	Delivery fee D. Up to and including 2.0 km: 15,000. Over 2.0 km up to and including 5.0 km: 15,000 + 5,000 × (whole km above 2, rounded up). Over 5.0 km up to 8.0 km: 30,000 + 7,000 × (whole km above 5, rounded up).	Rounded up means 2.01 km counts as 1 extra km
BR4	Rush-hour surcharge R. If the order is placed in [11:00, 13:00) or [17:30, 19:30), R = 20% of D. Start time is inclusive and end time is exclusive.	Uses the placement timestamp
BR5	Heavy-rain surcharge R2. While the Admin's heavy-rain flag is on, R2 = 10,000 flat. Otherwise 0.	Flag state is snapshotted on the order
BR6	Free delivery. If S is 300,000 or more, D and R are waived (set to 0). R2 still applies.	Boundary: 299,999 / 300,000
BR7	Quiet Kitchen discount Q. If hub load at placement is below 30%, Q = 10% of S. At exactly 30% there is no discount.	Boundary: 29% / 30%
BR8	Hub load and protection. Hub load = (orders currently in Confirmed or Preparing at that hub) ÷ (the hub's kitchen capacity, a whole number of concurrent orders). If load is 90% or above, the hub rejects new orders and the customer is asked to choose another hub.	The new order is not counted; boundary 89% / 90%
BR9	Vouchers. At most one voucher per order. A voucher must be active and within its validity dates. PERCENT type: V = p% of (S − Q), limited to the voucher's maximum discount. FIXED type: V = the fixed amount, but never more than (S − Q). The voucher's minimum-subtotal condition is checked against S.	Example: 10%, cap 30,000, min subtotal 100,000
BR10	Order of calculation. (1) S = sum of price × quantity. (2) Q per BR7. (3) V per BR9. (4) Food total F = S − Q − V. (5) D per BR3 and waiver per BR6. (6) R per BR4 (only if D is not waived). (7) R2 per BR5. (8) Raw total = F + D + R + R2. (9) Round per BR11.	Order matters; test it
BR11	Rounding. Round the raw total once, at the end, to the nearest 1,000 (half rounds up). Do not round intermediate values.	64,950 becomes 65,000
BR12	Order lines. Each line has quantity 1 to 20. An order has 1 to 30 lines. Only menu items marked Available can be ordered, and all items must belong to brands operated by the chosen hub.	Boundary: 0/1, 20/21, 30/31

ID	Rule	Notes
BR13	Payment methods by actor. Local Customer: MoMo, VNPay, Cash on Delivery. Foreign Tourist: Apple Pay, PayPal, Cash on Delivery. A customer cannot use a method that belongs to the other group.	Strategy pattern required
BR14	Cash on Delivery (COD). Allowed only if the rounded total is 1,000,000 or less. The shipper must confirm cash collected before the order can become Delivered.	Boundary: 1,000,000 / 1,001,000
BR15	Order lifecycle. An order follows the state machine in section C. Any transition not listed is invalid and must be rejected by the business layer, not just hidden in the UI.	Test invalid transitions
BR16	Cancellation and refund. A customer may cancel while the order is Placed or Confirmed. Kitchen Staff may reject an order while it is Placed, giving a reason. An online-paid order that is cancelled is refunded in full (simulated).	No cancellation after Preparing
BR17	Editing and price snapshot. Items may be edited only while the order is Placed. At placement the system stores a snapshot (timestamp, hub load, rain flag). Any recalculation after an edit uses that snapshot, not the current conditions.	Keeps price reproducible
BR18	Registration. Local: Vietnamese mobile number of 10 digits starting 03, 05, 07, 08 or 09. Tourist: passport number of 6 to 9 letters or digits, nationality, valid email. Both: unique identifier, password of at least 8 characters, stored hashed.	Same schema, different validation

C. Order Lifecycle (State Machine)

States: **Placed, Confirmed, Preparing, Ready, Out for Delivery, Delivered, Cancelled, Failed.** Delivered, Cancelled and Failed are terminal.

From	To	Allowed actor	Condition
Placed	Confirmed	Kitchen Staff	Hub of the order
Placed	Cancelled	Customer or Kitchen Staff	Kitchen must give a reason
Confirmed	Preparing	Kitchen Staff	–
Confirmed	Cancelled	Customer or Admin	–
Preparing	Ready	Kitchen Staff	–
Ready	Out for Delivery	Shipper	A shipper is assigned
Out for Delivery	Delivered	Shipper	COD: cash confirmed (BR14)
Out for Delivery	Failed	Shipper	Customer unreachable

> **Open question (deliberate)**
> How a shipper is assigned, and what happens financially after a Failed delivery, are **not** specified. Make a reasonable assumption, record it in your Ambiguity Log, and be ready to defend it.

D. Worked Example

A Local Customer orders items worth S = 200,000 from a hub at 20% load. The address is 3.2 km away, the order is placed at 15:00, the rain flag is off, no voucher is used, and the customer pays by COD.

Step	Calculation	Result	Rule
1	Item subtotal S	200,000	BR10
2	Q = 10% × S (load 20% is below 30%)	20,000	BR7
3	V (no voucher)	0	BR9
4	F = 200,000 − 20,000 − 0	180,000	BR10
5	D = 15,000 + 5,000 × ceil(3.2 − 2) = 15,000 + 10,000	25,000	BR3
6	R (15:00 is not a rush window)	0	BR4
7	R2 (rain flag off)	0	BR5
8	Raw total = 180,000 + 25,000	205,000	BR10
9	Rounded total (COD allowed: 205,000 ≤ 1,000,000)	205,000	BR11, BR14

E. Expected Core Entities (For CRUD Operations)

Your database and classes will likely need to manage (Create, Read, Update, Delete) the following entities:

- **Users / Actors:** Local Customers, Foreign Tourists, Kitchen Staff, Shippers, System Admins.
- **Hubs:** location, kitchen capacity, brands operated, current load (derived).
- **Brands & Menu Items:** brand, category (for example noodles, rice, drinks), price, availability.
- **Vouchers:** code, type (PERCENT or FIXED), value, maximum discount, minimum subtotal, validity dates, active flag.
- **Orders:** order lines, status and status history, price snapshot (BR17), payment record, assigned shipper, applied discounts and fees.

Where an entity is referenced by orders (hubs, menu items, users, vouchers), "Delete" should be a **soft-delete** (an inactive or deleted flag) so that history and reports stay correct. State this choice and justify it in your design.

F. Required Reports

Each group must implement **at least two** reports. The two below are the minimum; a third (for example on-time delivery rate) is optional.

- **R1 Revenue Report:** for a chosen date range, total revenue by brand category and by payment method. Cancelled and Failed orders are excluded.
- **R2 Hub Utilization Report:** for each hub, the number of orders handled, the average preparation time (Confirmed to Ready), the current load percentage, and the number of Quiet Kitchen discounts granted.

G. Multi-Gateway Payment Design

The system must structurally support the payment methods in BR13 using the **Strategy pattern**: one payment interface, one class per method, and a selection mechanism that depends on the actor type. Standard tiers may **simulate** the payments in the UI and database (for example a mock gateway with a documented interface). Tier 4 must call a live sandbox API.

3. Technology Tiers (Choose Your Target Grade)

You must self-select a technical tier. Your chosen tier dictates your technology stack and maximum base score. Choose a tier that challenges your team but remains realistic for a 7-week timeframe.

Tier	Max score	Stack and architecture	Guidance
Tier 1	7.0	C# WinForms + local SQL database / Excel. 3-Tier architecture.	Lowest setup risk; limited UI and deployment.
Tier 1B	7.5	Core PHP (or CodeIgniter) + MySQL + Bootstrap. MVC architecture.	Fast web start; you must enforce MVC yourself.
Tier 2	8.5	ASP.NET Web Forms/MVC (Database First) + SQL Server + Bootstrap. MVC architecture.	Good balance for most teams.
Tier 3	9.5	ASP.NET MVC (Code First) + Entity Framework + Bootstrap. MVC architecture.	Migrations and ORM add learning curve.
Tier 4	10.0	Tier 3 plus all three: (a) live sandbox API integration (VNPay Sandbox or PayPal Developer), (b) interactive Admin dashboard, (c) automated tests of core workflows (for example Selenium or Playwright).	Highest risk; partial completion stays within the Tier 3 cap.

Tier rules. (1) The instructor confirms your tier when Part 1 is marked. (2) Changing tier after Part 2 needs written approval. (3) The cap applies only if the delivered system really uses the stack of the tier (a mixed stack is graded at the lower tier). (4) The cap is a ceiling, not a guarantee: your mark is the rubric percentage, limited by the cap.

4. The 7-Week SDLC Roadmap & Deliverables

> **Traceability starts in Part 1**
> Give every requirement an ID (**FR1…, NFR1…**) and cite the IDs and the business rules (**BR1…**) in your use case specifications, diagrams, code comments and test cases. The RTM in Part 3 is then the final state of a table you have been keeping all along.

Part 1: Project Proposal & Requirements Engineering (2 Weeks)

Total Weight: 20% (15% Document + 5% Presentation)

The "1 Use Case Per Student" Requirement: Each student must take ownership of at least one main use case with **complete CRUD**. A group of 2 has 2 main use cases; a group of 3 has 3. Choose from the list

below or propose your own (with instructor approval). **"Place and Manage Orders" must be taken by one team member.**

#	Candidate use case	CRUD meaning
A	Manage Menu & Brands	Create, view, edit and remove brands and menu items; set availability.
B	Manage Hubs & Capacity	Create, view, edit and remove hubs; set kitchen capacity; view current load.
C	Manage Vouchers	Create, view, edit and deactivate vouchers with type, cap and minimum subtotal.
D	Place and Manage Orders (required)	Create an order with priced lines; view and track it; edit items while Placed; cancel when allowed. Includes the pricing engine and payment selection.
E	Manage User Accounts	Register Local Customers and Tourists with the correct validation; view, edit and deactivate accounts.
F	Manage Deliveries	Assign shippers; update order status along the lifecycle; handle failed deliveries; view delivery history.

The **pricing and payment module** (BR1 to BR14) is shared team infrastructure. The student who owns use case D is its named owner, and the others integrate with it through its public interface.

Document Checklist (what to include in your PDF):

1. **Title Page:** Project name, team members, student IDs, chosen tier.
2. **Project Proposal:** Chosen SDLC methodology (for example Scrum), team role breakdown, and a table stating which student owns which use case.
3. **Problem Statement:** A brief summary of the client's needs in your own words.
4. **Functional Requirements (FRs):** Numbered list ("The system shall…") with IDs, each mapped to your use cases, the relevant business rules and the 2 required reports.
5. **Quality Attributes (NFRs):** At least **three** specific, measurable attributes. The performance, maintainability, usability and security baselines in the case study are mandatory; add at least two of your own.
6. **Ambiguity Log:** At least three ambiguities, gaps or conflicts you found in this specification, the assumption you made for each, and the instructor answer if you asked one.

Presentation 1 (5%): A 5-7 minute "Solution Pitch" where all members speak and claim ownership of their use case.

Part 2: Software Design, Architecture & UI/UX (2 Weeks)

Total Weight: 25% (20% Document & Figma + 5% Presentation)

The "1 Use Case Per Student" Requirement: Students perform deep UML modelling and UI design for their individually assigned use cases.

Document Checklist (what to include in your PDF):

1. **Overall Use Case Diagram:** The system boundary with actor generalization (Customer, Local Customer, Foreign Tourist).
2. **Use Case Specifications:** For your chosen use cases: name, actor, pre-conditions, main flow (step by step), alternate/exception flows and post-conditions. Cite FR and BR IDs.
3. **Behavioral Diagrams:** A UML Sequence Diagram **or** Activity Diagram for each of your use cases, plus one team **State Diagram for the Order lifecycle** matching section C.
4. **Structural & CRC Design:** CRC cards to outline object interactions, followed by a formal UML Class Diagram (or ERD). The class design must show the **Strategy pattern** for payments.
5. **Architecture & Quality Design:** Visualize your 3-Tier or MVC setup. Explain how it applies quality design principles (SOLID, separation of concerns, high cohesion and low coupling) to satisfy your NFRs, and state where the pricing rules live and why.
6. **UI/UX Design:** Screenshots and a public link to your Figma prototype. Web tiers must be designed with a responsive grid (for example Bootstrap containers).

Presentation 2 (5%): A 7-10 minute "Design Review" walking through your CRC cards, database logic and Figma prototype.

Part 3: Implementation, Testing & Product Demo (2 Weeks)

Total Weight: 35% (30% Code/QA + 5% Presentation)

The "1 Use Case Per Student" Requirement: The codebase and QA testing must prove that each student's use case works according to the design, with complete CRUD.

> **Week 5 checkpoint and Change Request**
>
> At the start of Week 5, each team shows a **running skeleton** (database schema plus one use case end to end). The checkpoint is not graded, but a team that cannot show one meets the instructor.
>
> The instructor also issues a **Change Request** this week (for example a new payment method or a new pricing rule). A well-designed system absorbs it with small, local changes. Submit a **one-page Change Request Impact Note** listing the classes added and modified, with evidence that existing payment classes were not edited.

Deliverables Checklist:

1. **Source Code:** A tagged release in your GitHub repository is the official submission. A .zip copy is welcome as a backup. Web tiers must use Bootstrap.
2. **Version Control:** Link to your GitHub repository with **more than 10 meaningful commits per student**, plus the signed **Contribution Statement** (see section 6).
3. **Detailed Design & Justification:** Explain your final implemented design and compare it with your Part 2 design, justifying any structural, database or logic changes (or explaining why none were needed).
4. **QA & Testing Report (PDF):**

a) **Test Plan:** A table of test cases (Test ID, Description, Input Data, Expected Result, Actual Result, Pass/Fail) with the FR and BR IDs each case covers.

b) **Test Design Techniques:** Show explicitly how you used Equivalence Partitioning and Boundary Value Analysis for **every numeric threshold in BR1 to BR12 and BR14** (at least 20 EP/BVA test cases in total), and at least 5 tests that try **invalid order-state transitions** (BR15).

c) **Execution Evidence:** Screenshots of passed unit tests.

d) **(Tier 4 only):** Source code and execution logs of automated tests (for example Selenium or Playwright).

5. **Verification of Given Solution:** A Requirements Traceability Matrix (RTM) linking each FR and NFR to its use case, design element, code component and test cases, with a pass/fail verdict.
6. **Deployment:** A deployed app link (web) or the executable path (desktop). If hosting may expire after the deadline, also include a screen recording; the tagged repository and the video are the permanent evidence.

Presentation 3 (5%): A 10-15 minute live "Final Product Demo." Students execute their chosen use cases and show the 2 reports live. Each student answers at least one individual question about their own use case.

Final Submission: Comprehensive Report & Video Demo

Total Weight: 20% (10% Report + 10% Video)

Deliverables:

1. A single, consolidated **PDF report** of Parts 1, 2 and 3 with PDF bookmarks. You must apply the feedback the instructor gave you during the previous presentations.
2. **Lessons Learnt:** A dedicated section at the end where each team member reflects on their SDLC journey, the challenges they faced, and how they overcame them.
3. A polished **.mp4 Video Demo** (or unlisted YouTube link) walking through the working product end to end, including one order for each customer type and both reports.

5. Evaluation Rubric (100% Scale)

Note: Your final percentage is capped by your chosen Technology Tier (for example, a team scoring 100% at Tier 1 receives 7.0/10). Each row is marked on four levels: **Excellent (90-100% of the row weight), Good (70-89%), Adequate (50-69%), Weak (below 50%)**.

Item	Weight	Excellent	Good	Adequate	Weak
Part 1: Proposal & Requirements Doc	15%	FRs have IDs, are testable and map to use cases, BRs and reports. One CRUD use case per student is clear. At least 3 measurable NFRs. Ambiguity Log has 3 or more sensible assumptions.	Mostly testable FRs with IDs. Minor gaps in CRUD scope or NFR measurability. Ambiguity Log present but thin.	FRs listed but vague or unmapped. NFRs only partly measurable. Ownership unclear.	FRs copied from the spec without refinement. Reports or ownership missing. NFRs not measurable.
Part 1: Presentation	5%	Clear 5-7 minute pitch. All members speak and claim their use case. Ties the plan to client value. Handles questions well.	Clear pitch, all members speak, minor gaps in ownership or timing.	Pitch is hard to follow, or some members do not speak.	Unprepared, wrong length, or ownership not shown.
Part 2: Design Doc & Figma	20%	Complete use case specs, behavioral diagrams and Order state diagram. CRC and class diagram show Strategy pattern and clear separation. Architecture linked to NFRs. Figma matches the flows and reports exactly.	Most artefacts complete and consistent. Strategy pattern present but weakly justified. Minor Figma mismatches.	Several artefacts incomplete or inconsistent with each other. Quality principles named but not applied.	Missing major artefacts. No Strategy pattern. Figma unrelated to the design.
Part 2: Presentation	5%	Effective technical walkthrough of CRC, database logic and Figma. Prototype reflects multi-payment, load-aware pricing and the reports.	Good walkthrough with small omissions.	Walkthrough lacks depth or misses key requirements.	Cannot explain own design.
Part 3: Implementation & QA	30%	Chosen use cases with full CRUD work flawlessly. Strict MVC/3-Tier. Pricing rules isolated and correct. 20 or more EP/BVA cases and invalid-transition tests with traceable IDs. Design justification and RTM complete. Change Request absorbed with local changes (Tier 4: automated tests).	Use cases work with minor defects. Layers mostly separated. EP/BVA covers most thresholds. RTM mostly complete.	Core flows work but CRUD or separation is partial. EP/BVA limited. RTM superficial. Change Request needed wide edits.	Major features missing or broken. Business logic in UI. Few or no systematic tests.
Part 3: Live Demo	5%	Flawless live execution of chosen use cases and both reports. Each student answers individual questions confidently.	Works with small hiccups. Q&A mostly sound.	Partial demo or relies on screenshots. Weak Q&A.	Demo fails or members cannot explain their own work.
Final Comprehensive Report	10%	Parts 1-3 in one polished PDF with bookmarks. Previous feedback clearly addressed. Insightful, individual Lessons Learnt.	Complete and well formatted. Feedback mostly addressed. Reasonable lessons.	Parts stitched together with inconsistencies. Little evidence of feedback applied. Generic lessons.	Missing parts or bookmarks. Feedback ignored. Lessons missing or copied.
Video Demo	10%	High-quality, clear narration, end-to-end flow including both customer types and both reports.	Clear video with minor omissions in coverage or audio.	Hard to follow, or key flows missing.	Unusable audio or video, or product not shown working.

6. Individual Contribution & Peer Review

The team mark is adjusted for each member by a **Contribution Factor between 0.70 and 1.00**: individual mark = team mark × factor (then capped by the tier). The factor is based on four pieces of evidence:

- The **GitHub commit history** (more than 10 meaningful commits per student, and commits touching the student's own use case).
- A **Contribution Statement** signed by all members at the start of Part 3, stating who owns which use case, class group and report.
- A confidential **peer review** form submitted at the end of Week 6.
- **Individual questions** during Presentations 2 and 3.

The factor is reduced below 1.00 only when the evidence shows a clearly smaller contribution.

7. Declaration of Generative AI Use

The use of Gen AI tools (ChatGPT, Google Gemini, Claude, etc.) for brainstorming or refining language is permitted. However, all use must be transparent. Include an **Appendix.pdf** with a **Generative AI Usage Log** table:

Tool and date	Purpose	Prompt (summary)	What you kept or changed	How you verified it
e.g. ChatGPT, 20 Oct	e.g. draft FRs	e.g. "Write FRs for an order lifecycle…"	e.g. kept 4 FRs, rewrote 3 to add IDs	e.g. checked against BR table

Keep the **full prompts and outputs** in a separate Appendix B file; you must hand it over within 48 hours if the instructor asks for it. Failure to declare AI use will be treated as a breach of academic integrity.

8. A Note on the Importance of Presentations

While each presentation carries a smaller weight than the written reports or prototypes, they are crucial to your overall success:

- **Bridging the Gap Between Tech and Business:** As consultants, you translate technical work (state machines, pricing rules, database design) into clear business value for the client: fewer overloaded kitchens, clearer prices and happier customers.
- **Proving Authentic Understanding:** With Gen AI tools and group collaboration, your live presentation and individual Q&A are the strongest validation of your work. You must navigate your prototype and explain why you chose your data model and your design pattern.
- **Guiding the Assessor:** A clear presentation acts as a live executive summary. By highlighting your best design decisions, you tell the assessor what to look for in your documents.
- **Synthesizing the Project:** The project is deliberately split into requirements, design and implementation. The final presentation ties them together.

(Please refer to the separate "FP Student Survival Guide" for practical tips to get through the 6 weeks.) Good luck.